SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

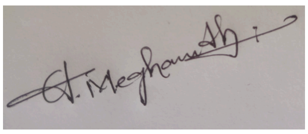
1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I _A.Meghanadh(192372281)___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

CSA43

CO-4 AT3

CLASS TEST

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1. Fundamental Concepts in Web development.

XML

(eXtensible Markup Language)

XML is a Markup language designed to store and transport data in a structured, human readable and machine-readable format. unlike HTML, which focuses on displaying data, XML focuses on describing data using custom-defined tags.

JSP (JavaServer Pages)

JSP is a server-side technology that allows developers to embed Java code within HTML pages using tags (<% %>). When a client requests a JSP page, the servlet.

JSP is used to create dynamic, data-driven web pages in Java based web applications.

MVC
(↳ Model-view-controller)
~~architecture~~

MVC is a diagram design pattern that seperate an application into three interconnected components.

* Model → Manages data and business logic
* View → Handles the presentation } UI layer
* Controller → processing user i/p and coordinates between Model and view.

- This seperation improves code organization, Maintainability, reusability and allows parallel development by different teams.

XSLT (eXtensible Stylesheet Language Transformations)

* XSLT is a language used to transform XML doc into other formats such as HTML, plain texts or another XML structure.

* XML Schema (XSD)

XML schema defines the structure, data types, and constraints of an XML document. It specifies what elements and attributes may appear.

JSTL

(JSP standard tag library)

JSTL is a collection of predefined tags that simplify JSP development by removing the record need for embedded Java.

PHP (Hypertext preprocessor)

PHP is a widely-used open-source server-side scripting language designed for web development. PHP code is embedded within HTML and executed on the server, generating dynamic content that is sent to the client's browser.

DATABASE CONNECTIVITY

Database Connectivity refers to the mechanism that allows a web application to communicate with a database to store, retrieve, update, or delete data. This is acheived through APIs such as JDBC (Java database Connectivity) for Java applications or PDO | mysqli for PHP applications, enabling dynamic, data-driven web content.

Parser

-* A parser is a program that reads and analyses a document (such as XML or HTML) to extract data and check its structure. In the context of XML, parsers (like DOM and SAX parsers) read the XML content and convert it into a format the application can use, while also validating it against a DTD or XML schema if required.

(2) Interconnection of these Technologies in a web Application.

* Modern web applications rarely rely on a single technology → they integrate these components together to acheive dynamic, structured and maintainable systems.

Structured and Validating XML

⇒ XML acts as the common language for data exchange. Data a is structured using custom XML tags.

* This structure is validating against an XML schema, which ensures the XML document follows the correct format, data types and hierarchy before it is processed further.

Server-side processing with JSP and PHP

When a user sends a request (submitting a form), it is handled by server-side technologies like JSP or PHP.

Organizing with MVC Architecture

* The model interacts with the database (using JDBC in Java or built-in functions in PHP) to fetch or updata data.
* The controller receives the client request, invokes the model to get/process data, and decides which view to render.
* The view displays the final output to the user.

DATABASE CONNECTIVITY AND DATA FLOW

DC (via JDBC, PDO etc) allows the model layer to execute SQL queries against the database - retrieving records, inserting new data or uploading existing entries.

Putting It all together - Exp flow

1. A user submits a request through a web form (view).
2. The controller (servlet/PHP script) receives the request

3. The controller calls the model, which uses database Connectivity to query the database.

4. Retrieved data may be structured as XML and validated using an XML schema.

5. A parser reads this XML, and XSLT may transform it for presentation.

6. The controller passes processed data to the view, built with JSP (using JSTL tags) or PHP, which renders the final dynamic HTML pages sent back to the user's browser.